

Assumptions

1. The **reporteurClass** (Pilot or Maintainer) in **tlbStats** is mutually exclusive, so a reporteur cannot be both.
2. If the **reporteurClass** is Maintenance Reporter (**MAREP**), the **airport** field must be **NOT NULL**; otherwise, it should be **NULL**.
3. All time variables representing minutes are stored as integers for simplicity.
4. **anyo**, **mes** and **dia** in the **time** table are accurately derived from the date field.

Our objective in this project is to simplify as much as possible the domain of the system. To do it we have centered our point of attention into the KPI's. Once we have understood how each KPI relates to the others we have established the three fact tables. The three tables we have defined are the next ones:

- **FlightStats** (identified by *aircraftId* and *flightDay*)

In that table we store the daily flight stats of each aircraft. In that table we store information related to the KPI's: *flight_hours* (FH), *departures* (TO), *cancellations* (CN) and *delays* (DY).

- **MaintenanceStats** (identified by *aircraftId* and *mainDay*)

In that table we store the scheduled and unscheduled maintenance events of each aircraft each day. In that table we store the variable *timeUnavailable* which is the time that an aircraft has been unavailable. Necessary for the calculation of the next KPIs:

- ADOS, ADOSU, ADOSS or ADIS.

- **TLBStats** (identified by *aircraftId* and *tlbDay*)

In that table we store the daily technical logbooks stats of each aircraft reported by a certain worker(pilot or maintenance worker). In that table we store the variable *logbookCount* which is necessary for the calculation of the next KPIs:

- RRh, RRc, PRRh, PRRc, MRRh, MRRc

We are conscious that our final schema requires some transformations during the ETL. During the ETL, we are going to calculate all the variables and load it into the database. We have considered our schema the best option because it maintains the information with the finest granularity, therefore occupies the least amount of storage possible.

In our model the dimensions related to each fact table are *aircraft* and *date*. In the *aircraft* dimension we have some variables like: *aircraftId*, *model* and *manufacturer*. We have conserved those variables because they are necessary for the calculation of some KPI's. On the other hand we have the *date* dimension, in which we only store the day, month and year. This is in fact a quasi degenerate dimension. We are conscious that with a variable with data type '*DATE*' we can extract that information, but in order to understand and clarify the dimensions, we have decided to create a table for days.

Lastly, usage of materialized views for caching or speeding up computations will be avoided, as the final schema already represents data at the smallest level of granularity plausible. Aggregated materialized views, even with a once-daily refresh, would add unnecessary overhead in terms of storage and compute resources. While a standard view could be useful for abstraction, KPI computations are beyond the scope of this project, so we will not include unmaterialized views either.